J. prakt. Chem. 337 (1995) 253–265

Journal für praktische Chemie
Chemiker-Zeitung
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Übersichtsartikel · Review Article

Electronegativity Effects in Organic Synthesis: Differences in Stability of Isomeric Substituted Alkanes

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Received December 21st, 1994

Dedicated to Professors Georges Leroy and Paul von Ragué Schleyer on the occasion of their 60th and 65th birthday

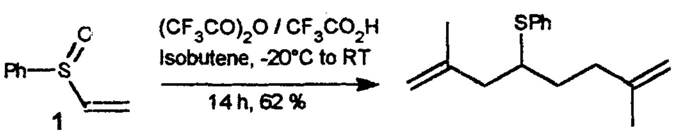
Abstract. Selectivity is such an intellectually, economically, and ecologically important matter in organic synthesis that it is highly desirable to know all the factors, both kinetic and thermodynamic, which influence it. We were surprised by the high selectivity obtained in an equilibration of isomeric β-thioalkyl trifluoroacetates, because the thermodynamic difference implied by this and other, similar, rearrangements did not appear to have been explained.

It now appears that this difference is due to the thermochemical effect whereby polar substituents bind more strongly to alkyl groups of complementary polarity; in other words electronegative atoms prefer more „procationic" tertiary and secondary alkyl groups, electropositive ones prefer the more „proanionic" primary alkyl and methyl groups. This small but significant (1 to 7 kcal·mol^{-1}) effect is well documented by thermochemical data, and has been described several times; it can be predicted qualitatively by Pauling's formula relating bond energy, bond polarisation, and electronegativity.

We first describe how the electronegativity effect explains our observations: they are due to the higher electronegativity of oxygen and chlorine relative to that of sulfur. We then go on to show that it is useful for the interpretation of many other literature reports, ranging from the acid-catalysed isomerization of alkyl halides, to the carbocyclization of organozinc reagents, and the isomerization of alkyl-transition metal complexes (a key step for the selectivity of important industrial processes such as alkene hydroformylation).

1. Introduction

The stability of isomeric substituted alkanes became a topic of interest in our laboratory following an unexpected observation made in the course of our studies on the chemistry of sulfur stabilised cations. In search of a synthetic equivalent of a vinyl α,β-dication, we investigated the Pummerer reaction [1] of phenyl vinyl sulfoxide **1** with various alkenes upon activation with trifluoroacetic anhydride [2, 3]. Only with isobutene, the desired double reaction was observed.



Scheme 1. Phenyl vinyl sulfoxide as a vinyl dication equivalent.

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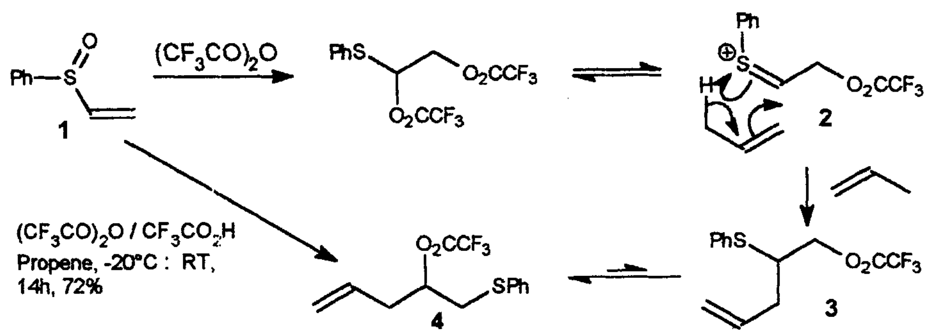
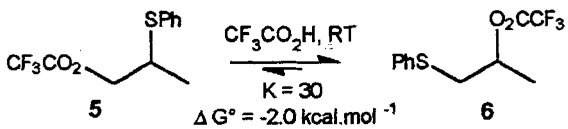
With other alkenes, such as propene, the β-thio trifluoroacetates such as **4** were obtained in good yield. Our mechanistic investigations [3] suggested that the initial, unobserved, product is the isomeric **3**, formed by an ene-type addition [4] to propene of the thionium ion **2**, itself derived *in situ* from phenyl vinyl sulfoxide **1**. As confirmed by the independent synthesis of compound **3**, it thereafter undergoes isomerization to **4** under the reaction conditions. (scheme 2)

We also synthesised the simpler analogues **5** and **6**, which equilibrate under the same conditions, with an equilibrium constant of about 30 in favour of compound **6**, in which the trifluoroacetoxy group is in the secondary position. At that time, we were not yet aware of an explanation for this significant difference in stability ($\Delta G° \simeq 2.0$ kcal·mol^{-1}). Steric effects alone could not be operative, since the steric bulk of the two groups, as measured [5] by their A values ($C_6H_5S^-$; 0.8 kcal·mol^{-1}; $CF_3CO_2^-$; 0.68 kcal·mol^{-1}) are similar. (scheme 3)



Scheme 2. Ene-type reaction of phenyl vinyl sulfoxide, and isomerization of **3** to **4**.



Scheme 3. Equilibration of simple β-thioalkyl trifluoroacetates.

2. Electronegativity

Electronegativity [6] has been one of the most enduring and popular concepts in chemistry, since it was introduced by Pauling, as *„the power of an atom in a molecule to attract electrons to itself.*" This *„power"* cannot be observed, but it seems to be a very fundamental property, since several very different methods which have been suggested for its determination, such as Pauling's bond-energy method; Allred and Rochow's charge/radius method; or spectroscopic measurements like those of Mulliken all give roughly similar scales. Electronegativity has been used to rationalise a large number of chemical properties.

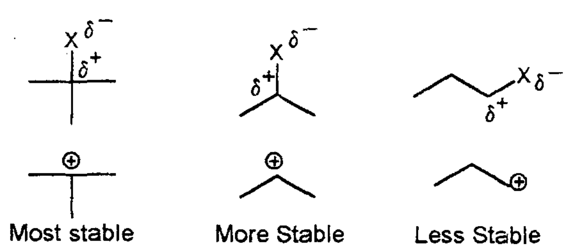
We now use it to account for the isomerizations described above, which we ascribe to the effect of substituent electronegativity on the stability of substituted alkanes. After summarising physico-chemical discussions of this thermochemical effect, we will explain how it provides the synthetically significant driving force for isomerization in our case, and in others drawn from the literature.

3. Stability of Isomeric Substituted Alkanes

In thermochemistry, it is a well acknowledged [7] experimental fact that alkanes substituted with an electronegative atom (or a substituent whose linking atom is electronegative, e.g. OH) in a secondary or tertiary position are more stable than their isomers, in which the substituent is on a primary carbon. The difference in stability is largest for the most electronegative substituents. This effect can be made apparent in several ways. The most obvious one is to calculate the difference $\Delta\Delta H_f^\circ$ in stability of isomeric substituted alkanes [8]. Thus, isopropanethiol is about 2.0 kcal·mol^{-1} more stable than n-propanethiol, whilst isopropanol is about 4.0 kcal·mol^{-1} more stable than n-propanol. tert-Butanol is as much as 6.9 kcal·mol^{-1} more stable than iso-butanol. It is also possible to compare the stability of non-isomeric substituted alkanes, for example of ethyl-X and methyl-X. Although the absolute value for this $\Delta\Delta H_f^\circ$ has no meaning in this context, the way it varies depending on the electronegativity of the substituent does. Luo and Benson [9] plotted $\Delta\Delta H_f^\circ$ (tert-butyl-X / methyl-X); $\Delta\Delta H_f^\circ$ (isopropyl-X / methyl-X); and $\Delta\Delta H_f^\circ$ (ethyl-X / methyl-X) against the Covalent Potential electronegativity [10] of substituent X. The trend is obvious: the stability of the branched compound relative to the methyl derivative always increases with increasing substituent electronegativity, and the increase is larger with the more highly branched alkyl chains.

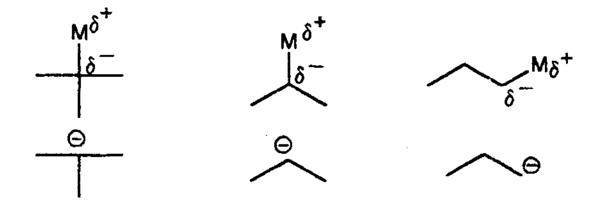
One can also calculate the bond separation energy [11] between a methyl group and a group X, defined as the energy of the process: $Me\text{-}CH_2\text{-}X + CH_4 \rightarrow Me\text{-}CH_3 + CH_3\text{-}X$. This process is unfavourable when X is electronegative, i.e. there is a stabilising interaction between geminal electronegative X groups and alkyl groups such as methyl. This highlights the fact that the electronegativity effects on substituted alkanes discussed here are really special cases of the more general σ-interaction effects between two substituents X and Y.

All the observed trends can be intuitively understood if one accepts that with increasing electronegativity of the substituent, the C-X bond is increasingly polarised, and the alkyl group is increasingly carbocation-like. The order of stability of the covalent compounds simply mirrors that of the corresponding carbocations (scheme 4).



Scheme 4. Intuitive rationalisation of stability differences of alkyl-X compounds.

For electropositive substituents, the situation is reversed. Although good thermochemical evidence is lacking for these organometallic species [12], theoretical studies [7f, 11] unambiguously show that primary isomers are more stable than secondary and tertiary ones. Here again, this effect can be seen to be related to the corresponding order of stability of carbanions (scheme 5).



Least stable Less Stable More Stable

Scheme 5. Intuitive rationalisation of stability differences of organometallic compounds.

Schleyer has called [13] these differences in stability of isomeric substituted alkanes *sigma effects*, since they are due mainly to the inductive effect [14] on the C-X, C-C, and C-H sigma bonds. Accordingly, heats of formation of substituted alkanes can be predicted with high precision using a simple polarisation and charge interaction model [15].

It should be emphasised that the comparison with ionic species is only an intuitive model, based on the presumption that it is the polar, carbon-element bonds which lead to the differences in energy from one isomer to another. It is however extremely difficult to assign energies to individual bonds in a molecule. Energy is a property of the whole molecule, not of particular parts of a Lewis structure – the drawing of which is a matter of convention, since the true wave function of a molecule does not neatly separate electrons into bonding pairs.

Two types of bond energies can be defined; bond energies and bond dissociation energies (BDEs). These last do not give a precise figure for the bond energy in a multiatomic molecule. They are the energies of a process, and therefore depend both on the stability of the starting molecule and of the resulting radicals. Isomeric compounds will lead to isomeric radicals, the difference in BDEs is a result of the difference in stability of the isomeric substituted alkanes RX and of the resulting alkyl radicals R. Therefore, if one wishes to use this difference to assign bond energies in the starting molecules RX, one must first have assigned them in the radicals R, which is likely to be more difficult to do than with the closed-shell molecules!

Bond energies in molecules are really bond energy *increments*,[2] constructed so that their sum for whole molecules will be as close as possible to the experimental atomisation energies. In other words, these atomisation energies for a series of reference compounds are „separated" in a conventional way, and on the basis of various assumptions, into individual bond contributions. Although the assumptions aim to be as chemically plausible as possible, there is no reason to believe that the resulting increments fully mirror the underlying chemical reality. For example, some schemes of parameters for the heat of atomisation of alkanes include fairly different energies for the C-H bond in CH_3, CH_2 and CH groups, decreasing in that order. This „explains" why branched alkanes are more stable than linear ones, since they have more hydrogens in methyl groups.

However, this difference between E(C-H) values only occurs because the E(C-C) is assumed to be constant in alkanes. Furthermore, it is argued that C-H bonds in methyl groups are shorter and have higher BDEs than those in methylene or methyne groups, and should therefore be stronger. The difference in BDEs in clearly due to the order of alkyl radical stability; whereas the small difference in bond lengths is inconclusive: it could also be used as an argument for assigning different E(C-C) values, since rC-C decreases upon going from ethane to propane to isobutane.[3]

It should be remembered that bond energy increments have *no separate, intrinsic* physical meaning, since they are a result of the assumptions (which must be arbitrary !) used to define them. Plausible assumptions lead to plausible individual bond energies, but not to exact ones, and the validity of the bond additivity scheme should not be taken as evidence in favour of the assumptions. For example, the four-parameter Laidler-type scheme mentioned above assumes that all C-C bonds in alkanes have the same energy, and therefore assigns three different C-H bond energies to C-H bonds. However, a group additivity scheme (GAS) with the four parameters [C-(C)$(H)_3$], [C-$(C)_2(H)_2$], [C-$(C)_3(H)_1$], and [C-$(C)_4$], can be shown to be *mathematically identical*, for certain values of the parameters, to the bond energy increment scheme, i.e. they both predict the same heat of atomization for all alkanes. The GAS does not explicitly assign energies to C-C or C-H bonds, it can therefore be understood either as assigning different C-H bond energies with a constant C-C bond energy (Laidler), *or* as assigning a *constant* E(C-H), with *different* E(C-C)s! Thus no argument for different C-H or C-X bond energies can be drawn from bond additivity schemes.[4]

[2] For a discussion of these systems, see refs. [7e], [8] and [21].

[3] R. D. Topsom, in *Prog. Phys. Org. Chem.*, vol. 16 (Ed.: R. W. Taft), Wiley-Interscience, **1987**, p. 85–124. In fact, the bond length in ethyl-X decreases with increasing electronegativity of X (H, CH_3, Cl, Br, OH, F), which is suggestive as to how the electronegativity effects might act.

[4] See ref. [8], p. 531–560. Consider the three schemes: a) E(C-C) = 85.48, $E(C\text{-}H)_p$ = 98.19, $E(C\text{-}H)_s$ = 97.27, $E(C\text{-}H)_t$ = 96.53; b) [C-(C)$(H)_3$] = 337.32, [C-$(C)_2(H)_2$] = 280.02, [C-$(C)_3$(H)] = 224.75, and [C-$(C)_4$] = 170.96; and c) E(C-H) = 99.00 (arbitrary),

The model whereby R-X compounds are likened to carbocations, and R-M compounds to carbanions readily accounts for the relative stability of their isomers. Although it is undoubtedly very naive, this model provides the best explanation of the contrast between the RX and RM compounds.

4. A Theoretical Interpretation

It is interesting to note that more or less linear correlations have been obtained [7d, 9] between the enthalpy terms $\Delta\Delta H_f^\circ$ (RX/QX) and the electronegativity of the substituent X, where R and Q are two alkyl groups. As it has been suggested before [16], this linear relation can be easily predicted from Pauling's definition of electronegativity, eq. (a)

$$E_{AB} = 23(\chi_A - \chi_B)^2 + (E_{AA} + E_{BB})/2 \qquad \text{(a)}$$

Where χ_A and χ_B are the electronegativities of the bonded atoms (or groups) A and B, and E_{AB}, E_{AA}, and E_{BB}, the energies of the respective bonds. For an exchange reaction (b) involving four atoms or groups Q, R, X and Y, the enthalpy[5)] change is given by equation (c):

$$QX + RY \rightarrow QY + RX \qquad \text{(b)}$$

$$\Delta H^0_{(b)} = E_{QX} + E_{RY} - E_{QY} - E_{RX} \qquad \text{(c)}$$

Using equation (a), this can be expressed as a function of the electronegativities:

$$\Delta H^0_{(b)} = 46(\chi_R - \chi_Q)(\chi_X - \chi_Y) \qquad \text{(d)}$$

Taking Y as a reference substituent, this becomes a general equation (e):

$$\Delta\Delta H^0_f(RX/QX) = 46(\chi_R - \chi_Q)\chi_X + \text{constant} \qquad \text{(e)}$$

This theoretical equation predicts linear relationships between $\Delta\Delta H_f^0$ and the electronegativity of X, as has been empirically observed. Furthermore, the sign and relative magnitudes of the predicted slopes agree with experimental ones providing that the group electronegativities of alkyl groups are in the intuitively expected order $\chi_{Me} > \chi_{primary} > \chi_{secondary} > \chi_{tertiary}$.

For several reasons, though, equation (e) cannot be used quantitatively for the prediction of enthalpies for reactions in which substituted alkanes undergo isomerization or alkyl exchange. First, it is difficult to obtain reliable group electronegativities for alkyl groups or other multiatomic substituents. For example, a popular method for calculating group electronegativities, the electronegativity equalisation procedure, gives the same values of χ for isomeric groups such as n- and *iso*-propyl [6]. For the alkyl-transition metal organometallics discussed below, it appears even more difficult to get significant values for the electronegativity of the substituent, i.e. a transition metal with its ligands.

Second, the equation is probably too simple and therefore incorrect. Thus, although some methods do lead to different χ values for different alkyl groups, in the expected intuitive order mentioned above [17], inserting these values in equation (e) leads to slopes very different from the empirically observed ones. Apparently, it is a crude approximation of the Pauling equation (a) to assume a constant coefficient equal to 23. Theoretical studies [18] suggest that it should be instead a complicated function dependent on the atoms considered.

However, qualitative predictions are easy to make, on the basis on the following very simple rules applied to substituted alkanes, R-X: (α) The more the group X draws electrons to itself, the more it will prefer to bind to a procationic secondary or tertiary alkyl group R. (β) The more the group X releases electrons, the more it will prefer to bind to a primary (or methyl) alkyl group R. (γ) The degree of polarisation of the R-X bond can be assessed in simple cases by considering the electronegativity of the binding atom in X (see table 1), in more complicated cases by taking into account substituent groups (ligands) on this binding atom, its oxidation state, etc.

5. Selectivity from Electronegativity

The quite substantial enthalpic differences (1–7 kcal·mol^{-1}) discussed above can contribute significantly to the ΔG° of reactions in which a polar group bonded to one sort of alkyl group in the reactants becomes attached to a different one in the products.[6)] We hereby

[4)] Continued
$E(C_x\text{-}C_y) = E(C_x) + E(C_y)$, where $E(C_p) = 40.32$, $E(C_s) = 41.01$, $E(C_t) = 41.92$ and $E(C_q) = 42.74$. (all energies in kcal·mol^{-1}). The last scheme assumes a constant, arbitrary value for E(C-H), and different E(C-C) depending on the substitution of both carbon atoms, thus $E(C_p\text{-}C_s)$ in propane = 81.33, $E(C_t\text{-}C_s)$ in 2-methylbutane = 82.93. The three schemes predict completely identical heats of formation for all alkanes.

[5)] This should be internal energy E rather than enthalpy H, but for our purposes $\Delta H = \Delta E$.

[6)] The ΔG° of such reactions well of course depend on the entropy of isomerisation. For symmetry reasons, this entropy must be taken into account to obtain correct values for ΔG° (although it does not reverse the order of stability) with simple compounds such as i-PrX and n-PrX (see ref. [7d]). For the more complicated, less symmetrical compounds, discussed below, the entropy can be presumed to be roughly similar for isomeric compounds, and does not therefore influence our qualitative considerations.

Table 1 Pauling Electronegatities of Selected Atoms.

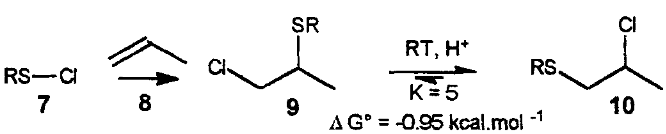
H	2.1	C	2.5	Na	0.9	Cl	3.0
Li	1.0	N	3.0	Mg	1.2	Br	2.8
Be	1.5	O	3.5	Al	1.5	I	2.5
B	2.0	F	4.0	S	2.5		

review examples of these reactions, which clearly show that selectivity in organic synthesis can be caused by these electronegativity effects. To start with, we return to the case of the β-thioalkyl chlorides and esters which first aroused our interest.

5.1. Non-metal Substituents.

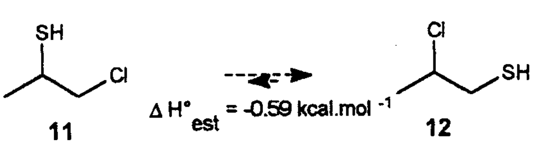
5.1.1. β-Thioalkyl Chlorides

The addition of sulfenyl chlorides **7** to alkenes (such as propene **8** for example) leads to mixtures of isomeric β-thioalkyl chlorides **9** and **10**. The mechanism of this reaction has been extensively discussed [19], and is usually assumed to involve a three-membered ring intermediate, whether an episulfonium ion or a (more or less) covalent σ-sulfurane. Isomerization of **9** to **10** takes place via the same intermediate. It has been known for some time [20] that the „Anti-Markovnikov" isomer **9** is the kinetic product, which isomerises on heating, or under acid catalysis at room temperature, to a mixture (usually about 85/15) of the more stable „Markovnikov" isomer **10** and of **9**. To our knowledge, however, no explanation has been given for this greater stability. (scheme 6)



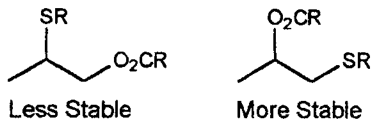
Scheme 6. Formation and isomerization of thioalkyl chlorides.

The explanation is that chlorine (3.0) is more electronegative than sulfur (2.5). The „preference" of the chlorine for the secondary position is greater than that of the alkylthio group. We are able to estimate the ΔH° for these isomerizations by adding group additivity values [21] (GAV's) for the model compounds **11** and **12** (scheme 7). The result is similar to the observed ΔG°.



Scheme 7. GAV estimate for model isomerization.

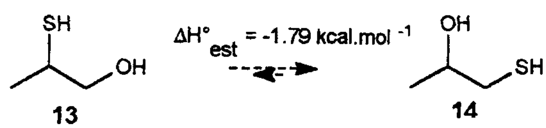
5.1.2. β-Thioalkyl esters



Scheme 8. Relative stability of β-thioalkyl esters.

Isomeric β-thio esters, formed either from β-thio alcohols [22] or from addition of sulfenyl esters to alkenes [23], can be equilibrated, via episulfonium ions or some similar species, by heating or under acid conditions. In all cases, the isomer in which the acyloxy substituent is on the more substituted carbon is strongly favoured (schemes 2, 3 and 8).

This can again be explained by the much higher electronegativity of oxygen (3.5) relative to sulfur (2.5). We can also estimate the ΔH° for the equilibria of schemes 2 and 3 by calculating the ΔH° between model compounds **13** and **14**. The estimated ΔH° is in line with the observed ΔG° (scheme 9).



Scheme 9. GAV estimate for model isomerization.

The difference of electronegativity between oxygen and sulfur is larger than that between chlorine and sulfur, so the equilibrium constants are larger with β-thioalkyl esters. This means that selectivity can be very high, and is a clear advantage of these oxygenated compounds over the corresponding chlorides. Fluorine is even more electronegative than oxygen, so it would be interesting to study isomerizations involving it, for example, the still undescribed [24] equilibria of β-thioalkyl fluorides [25].

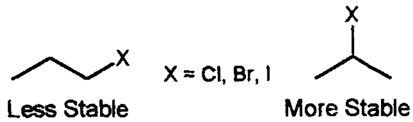
5.1.3. Alkanes

In the presence of strongly electrophilic catalysts, *n*-alkanes isomerise to give mixtures containing mainly the more stable branched isomers [26].

5.1.4. Acyl Chlorides

The chlorides of linear carboxylic acids such as n-octanoyl chloride isomerise at 130° in the presence of Pd (II) salts to give mixtures containing mainly branched acyl chlorides such as α-ethyl hexanoyl chloride [27].

5.1.5. Alkyl Halides



Scheme 10. Relative stability of alkyl halides.

The Markovnikov rule for electrophilic addition to unsymmetrical double bonds has a kinetic basis: addition, of hydrogen halides for example, proceeds through the more stable carbocation. As mentioned above, however, the stability of alkyl-X compounds mirrors that of the corresponding carbocations, and addition also leads to the more stable isomer (scheme 10). Under rather hard conditions, primary alkyl halides may equilibrate to mixtures containing mainly the secondary isomer. For example, *n*-propyl chloride and bromide isomerise to the corresponding *iso*-propyl derivatives either at high temperature, or at room temperature in the presence of aluminium halide [28 ,29] or under radiation [30]. The iodides also undergo isomerization by photolysis in solution [31].

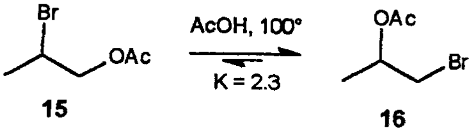
Addition of chlorine or bromine to *p*-allylanisole leads to mixtures of dihalopropanes $ArCH_2CHXCH_2X$ and $ArCH(CH_2X)_2$. The chlorides have been equilibrated [32], the equilibrium constant at 140 °C is of 9 in favour of the more stable secondary chloride.

5.1.6. Alcohols

Under highly acidic conditions, *n*-propanol isomerises to yield almost only *iso*-propanol [33].

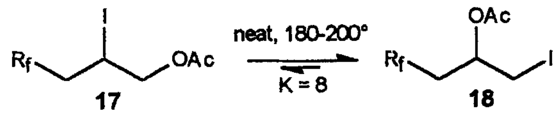
5.1.7. Haloalkyl Esters

Both pure bromoacetoxypropanes **15** and **16** lead to an equilibrium mixture containing mostly **16** upon heating for 30 h in glacial acetic acid [34]. This is expected, since oxygen is more electronegative than bromine. (scheme 11)



Scheme 11. Equilibration of bromoacetoxypropanes.

Similar rearrangements have been reported for dihalopropyl-acetates [35] or methyl carbonates [36]. The iodo perfluoroalkyl acetates **17** and **18**, obtained from radical addition of perfluoroalkyl iodides to allyl acetate [37] also lead to an identical mixture from either of the pure isomers (scheme 12).



Scheme 12. Equilibration of iodo perfluoroalkyl acetates.

5.2. Organometallic Compounds

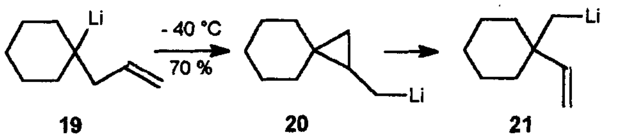
5.2.1. Alkyllithium Compounds

It is well known that the common organolithium reagents are of decreasing reactivity in the order *t*-BuLi > *sec*-BuLi > *n*-BuLi > MeLi. This chemistry is complicated by the question of aggregation [38], often held to be the main cause of differences in reactivity. However, theoretical studies [7f,11,39] (i.e. for monomers in the gas phase) clearly show the same trend: the stability of alkyllithiums is in the order Me > 1^{ary} alkyl > 2^{ary} alkyl > 3^{ary} alkyl. Also, the thermochemical data for the reaction between RLi and isopropyl alcohol in 90:10 hexane:ether have been obtained [40]. The heats of reaction correlate remarkably well with the ones computed from the theoretical data, thus indicating that the heats of aggregation for the various alkyllithiums are probably very similar.

Other experimental evidence comes from the 1,2 vinyl transposition of homoallyllithiums [41]. It is notable that this isomerization always involves complete conversion of secondary or tertiary alkyllithiums to less substituted primary ones. In this reaction, a vicinal lithium atom and vinyl group formally exchange positions, under very mild conditions. For example [42], the tertiary lithium derivative **19** rearranges, through 3-exo-trig ring closure to give an unstable cyclopropylmethyllithium **20**, followed by ring opening to yield the stable isomeric primary alkyllithium, **21**. (scheme 13)

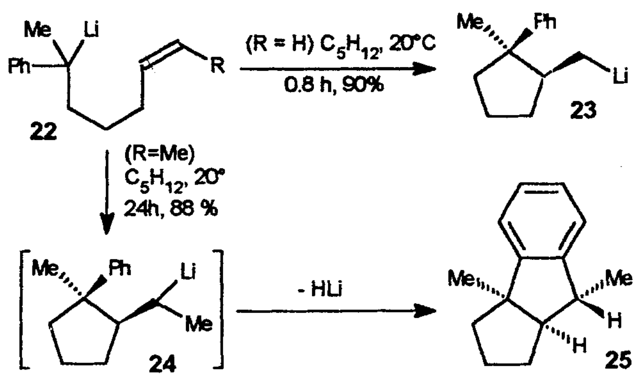
The relative stability of organolithium species is also apparent from their differing reactivity during analogous *irreversible* 5-exo-trig cyclization of 5-hexenyllithium derivatives. Thus, primary alkyllithiums undergo intramolecular addition at room temperature to a terminal C-C double bond in position 5 to give *primary* cyclopentylmethyllithiums [43]. If the double bond is 1,2-dialkyl-disubstituted [44], i.e. if the incipient lithium compound is in a less stable secondary position, cyclisation does not occur[7]. When the starting lithium compound is more reactive, as in the benzylic tertiary alkenyllithiums **22**, cyclization occurs in both cases [45], leading respectively to the primary derivative **23** and

[7] Except if the distal substituent on the double bond is able to stabilize a negative charge (phenyl, trimethylsilyl, cyclopropyl). See ref. [43].



Scheme 13. 100 % conversion of a tertiary alkyllithium to a primary alkyllithium.

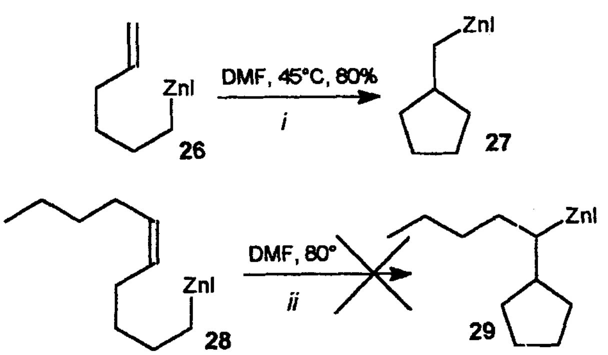
the secondary one **24**. This last compound, however, is reactive enough to further cyclise onto the phenyl ring, leading to **25**. (scheme 14)



Scheme 14. Contrasting behaviour of primary and secondary alkyllithiums.

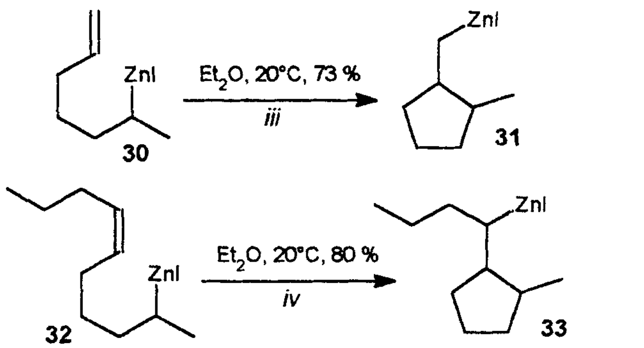
5.2.2. *Alkylzinc Compounds*

The pattern of reactivity in intramolecular carbometallations is even more clear with the corresponding organozinc reagents [46]. Thus, primary alkylzinc derivatives such as **26** cyclise at 45 °C in DMF (or at room temperature in ether) to give the primary cyclopentylmethyl zinc **27**, but the compound with the disubstituted double bond, **28**, does not lead to the less stable secondary zinc compound **29**, even at 80 °C. (scheme 15)



Scheme 15. Intramolecular carbometallation of primary alkylzinc derivatives.

The behaviour of the corresponding, more unstable, *secondary* organozinc reagents is in sharp contrast. Thus, the secondary reagent **30** leads to the corresponding cyclic primary alkylzinc **31**, and the disubstituted compound **32** also cyclises, even though the resulting alkylzinc **33** is secondary and thus less stable (scheme 16). The four reactions (i, ii, iii, iv) cover all the combinations of primary and secondary starting compound and cyclised product. It is striking that the only unsuccessful reaction is ii (scheme 15), the only one which would lead from a more stable primary compound to a less stable secondary one.



Scheme 16. Intramolecular carbometallation of secondary alkylzinc derivatives.

5.2.3. *Grignard Reagents*

Grignard reagents such as *sec*-butyl magnesium chloride are converted to primary isomers in the presence of $TiCl_4$ or $Ti(OR)_4$ as catalyst [47]. Complete conversion is rather difficult to obtain due to degradation, but ratios as high as 9 to 1 have been observed [48]. 1,2-Vinyl rearrangements such as described above for lithium compounds also occur with Grignard reagents [49], albeit at higher temperature (ca 80 °C, instead of −40 °C to RT). Here again, secondary organomagnesiums are converted selectively to more stable primary ones [50]. Also, exchange reactions between dialkylmagnesium compounds and dialkylmercury compounds show that primary alkyl groups much prefer to bond to magnesium than to more electronegative mercury [51].

5.2.4. *Boranes*

Trialkylboranes isomerise at high temperature to give mixtures in which primary boranes predominate [52]. Multiple isomerization can even occur, whereby the boron migrates along a long chain to reach the terminal position. Using the extensive chemistry of boron compounds, the tri-n-alkyl borane then obtained can be used for various synthetic purposes. The overall sequence of hydroboration, isomerization, and functional group interconversion is a useful method for the selective transformation of internal alkenes into terminally functionalized compounds, such as alcohols, for example [53].

Steric effects are usually postulated to act as the driving force for these reactions, and undoubtedly contribute in part to the large observed equilibrium constants. For example, any of the boranes **34**, **35**, or **36** lead to the same equilibrium mixture upon heating in diglyme [54]. The primary isomers **34** and **36** are largely favoured over the much more hindered secondary isomer **35**, and compound **34**, in which the alkyl group is branched in the β-position, is slightly more stable than **36**, in which the branching is in α (scheme 17).

However, electronic effects are also important, as has already been suggested [13]. For example, the diodo monoalkyl borane **37**, in which there is less steric strain, also isomerises [55] to give mainly the primary isomer **39**, with smaller amounts of the secondary isomers **37** and **38** (scheme 18).

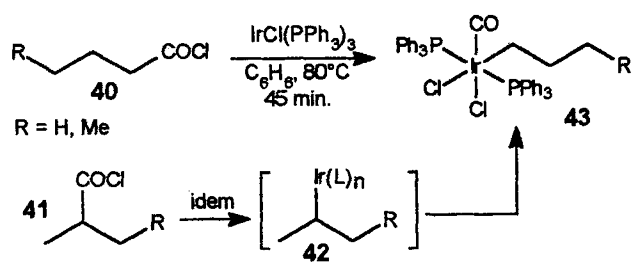
Likewise, the equilibrium ratio of $(n\text{-Pr})_3B$ to $(i\text{-Pr})(n\text{-Pr})_2B$ at 146 °C is of 6 to 1, i.e. there is a $\Delta G°$ for this reaction of 1.5 kcal·mol^{-1}, too large to be of steric origin alone [56]. It should also be mentioned that CH_3BH_2 is calculated [11] to be 1.5 kcal·mol^{-1} more stable than $C_2H_5BH_2$.

5.2.5. *Alanes*

Similar isomerizations have been reported [57] for the industrially important alkylalanes. For example, triisobutylaluminium reacts at 110 °C with internal alkenes, such as mixtures of *n*-octenes, to give mixtures of isomeric trioctylaluminiums with loss of isobutene. When catalytic amounts of transition metal salts such as titanium tetrabutoxide are added, the reaction is faster, and leads to tri-1-octylaluminium in greater than 98 % selectivity [58].

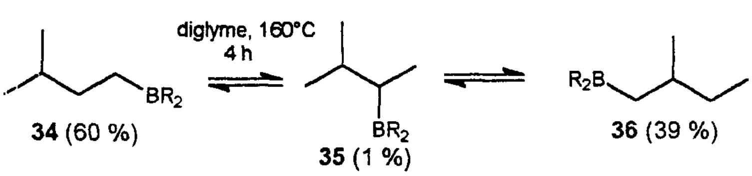
5.2.6. *Transition Metal Alkyl Complexes*

Secondary or tertiary alkyl complexes undergo isomerization towards primary alkyl isomers. This process is actually easier than with alkyl derivatives of main-group elements, due to the existence of a relatively low-energy mechanism, involving β-elimination to form an hydridometal alkene complex, followed by regioisomeric reinsertion. This can take place under rather mild conditions, so that a synthesis sometimes leads to a primary alkyl complex instead of the expected tertiary or secondary alkyl one, which has presumably rearranged immediately under the reaction conditions. In the example [59] shown in scheme 19, the two isomeric acyl chlorides **40** and **41** both lead to the same primary alkyl complex **43** after isomerization of the presumed intermediate **42**.

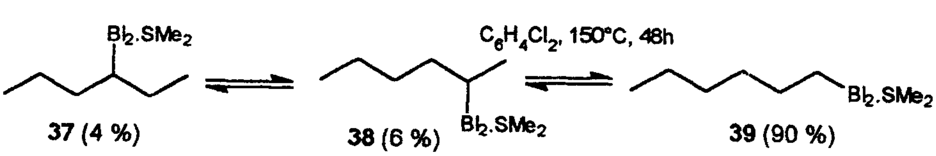


Scheme 19. Formation and *in situ* isomerization of the alkyl complexes **42** and **43**.

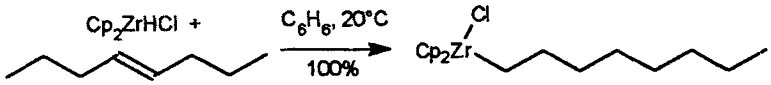
Hydrozirconation [60] of alkenes under mild conditions with $[(Cp)_2ZrHCl]$ (Cp = C_5H_5) leads to alkylzirconium compounds $[(Cp)_2ZrRCl]$ in which the R group is primary, after several isomerization steps (scheme 20). Subsequent electrophilic cleavage with various reagents [61] leads to terminally functionalized alkanes. Like the hydroboration-isomerization-functionalization sequence mentioned above, this is a contrathermodynamic process in that more stable inner alkenes lead to terminal compounds apparently derived from less stable terminal alkenes.



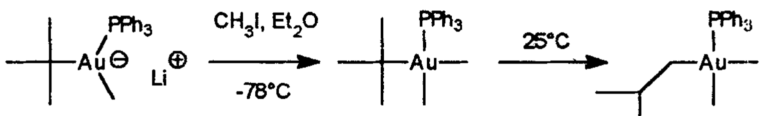
Scheme 17. Equilibration of tri(methylbutyl)borane.



Scheme 18. Equilibrium of diodo boranes.



Scheme 20. Formation of the more stable primary alkyl zirconium complex.



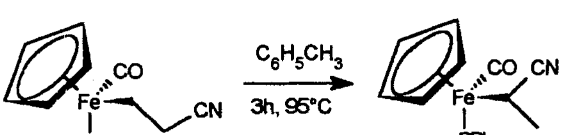
45 44 46

Scheme 21. *tert*-Butyl to *iso*-butyl rearrangement in gold complex **44**.

Other secondary and tertiary alkyl complexes can be isolated, but undergo thermal equilibration. For example [62], the gold (III) complex **44** is formed at −78 °C from the anion **45**, and detected by spectroscopy. Even careful attempts to purify it, however, lead to isomerization to the iso-butyl compound **46** (scheme 21).

In the 18 electron Iridium (III) complexes [IrXIR-(CO)(L)$_2$] (X = Cl or I, L = PMe$_3$ or PMe$_2$Ph), the *sec*-butyl and *iso*-propyl ligands are transformed to *n*-butyl- and *n*-propyl ones, respectively, under conditions favouring dissociation of a halide, and formation of a 16 electron complex which can then undergo β-elimination [63]. It is noteworthy that the equilibrium is completely displaced, even with the very small trimethylphosphine ligand, which argues against the sole action of steric effects.

Many other examples have come from the laboratory of Reger et al. The iron complex [CpFe(CO)(PPh$_3$)(*sec*-butyl)] quantitatively isomerises [64] at 60 °C over four hours to give [CpFe(CO)(*n*-butyl)]. However, the equilibrium in the case of the corresponding isomeric cyanoethyl complexes **47** and **48** lies completely in the other direction [65], as shown in scheme 22. The Fe-C bond is polarised, placing a partial negative charge on the carbon atom, which is of course much favoured next to the electron-withdrawing nitrile group. This electronic effect completely overweighs the steric effect.



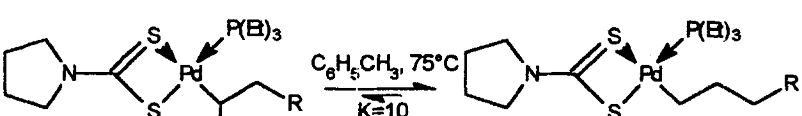
47 48

Scheme 22. Isomerization towards a branched alkyliron complex **40**.

Reger's group has also synthesised a set of very stable square planar palladium or platinum alkyl complexes such as **49** and **50**. The alkyl isomerization of these compounds is relatively easy to observe (scheme 23), and the x-ray crystal structures [66] establish that there is very little steric hindrance from the ligands. Nevertheless, the equilibrium is strongly displaced (K_{eq} is usually in the range 9–12) in all cases, towards the primary alkyl isomer [67].

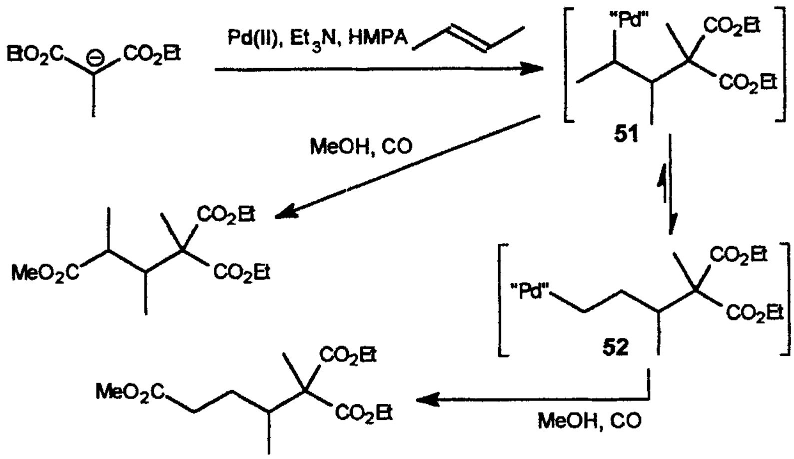
Isomerization of secondary or tertiary alkyl complexes sometimes occurs in an unexpected or unwanted way during other reactions. The coupling of *iso*-propyl magnesium chloride with aryl halides catalysed by [NiL$_2$Cl$_2$] (L = various R$_3$P) leads to varying proportions of *n*- and *iso*-propyl arenes, due to equilibration of the intermediate [RNiArL$_2$] species [68]. Carboacylation of 2-butene leads to mixtures due to competitive rearrangement [69] of the alkyl complex **51** to **52** (scheme 24).

The higher stability of primary alkyl complexes is also apparent in activation of alkane C-H bonds by transition metals [70]. Some coordinatively unsaturated complexes are able to undergo oxidative addition of alkanes to give alkyl hydrido complexes. Various studies have shown that insertion into a primary C-H bond is favoured kinetically and thermodynamically [71]. The reaction shown in scheme 25 has a characteristically high selectivity [72].

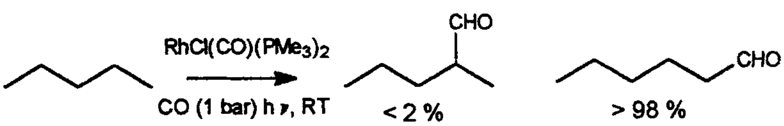


49 (R = H, Me) 50

Scheme 23. Equilibrium of alkyl palladium complexes.



Scheme 24. Isomerization of alkyl complexes during carboacylation.

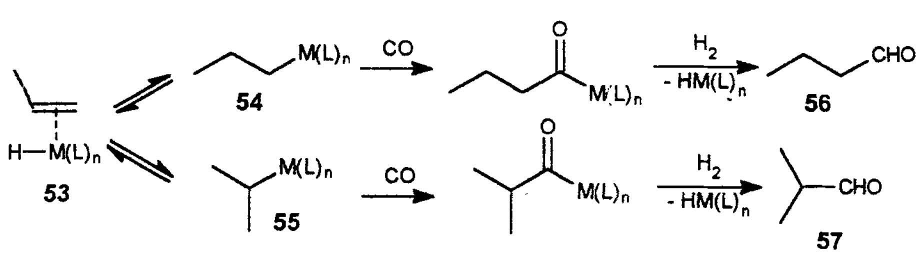


Scheme 25. Selective Carbonylation of Pentane.

The oxo hydroformylation process [73] of alkenes with carbon monoxide and hydrogen is an important industrial process, which is catalysed by many organometallic compounds, especially rhodium and cobalt carbonyl and their derivatives. The catalytic cycle is summarised in scheme 26. The alkene forms a π-complex **53** with the catalyst, which undergoes reversible insertion to give the *n*-propyl or *iso*-propyl complexes **54** or **55**. These species then lead irreversibly to the corresponding acyl complexes, and thence by hydrogenation, to the desirable linear aldehyde **56** or its branched isomer **57**.

In most cases, both types of insertion take place, and mixtures are therefore obtained, in which the linear compound **56** is the major component. Providing that the insertion and elimination reactions that lead to the equilibria between **53**, **54** and **55** are fast[8] relative to the insertion of CO, and that this insertion takes place at similar rates from the two complexes, the reaction is under partial[9] thermodynamic control, in that the observed ratios of **56** and **57** reflect the relative stability of **54** and **55**. Since we expect the primary alkyl complex **54** to be the more stable one, it is therefore understandable that hydroformylation leads mainly to linear aldehydes such as **56**.

It is of practical importance to increase the *n*-/*iso*-selectivity in this process. Although many other factors have been discussed as influential on this selectivity [74], we suggest that the electronic nature of the ligands should also be considered. More electron-donating ligands could increase the polarisation of the metal-carbon bond, and thus increase the partial negative charge on carbon, favour the primary alkyl complex **54**, and thereby increase the *n*-/*iso*-selectivity. It is known that metal carbonyl-based catalysts, in which back-bonding to the π^*_{CO} orbital reduces the electron density on the metal, give fairly low selectivities. When more electron-donating phosphines replace one or more of the CO ligands, selectivity increases [73, 75]. This is usually consid-



Scheme 26. Catalytic cycle in hydroformylation of propene.

[8] In many cases, isomerization of the alkene (through **54** and **55**) takes place faster than hydroformylation.

[9] Because C is more electronegative than H, the branched compound **57** is more stable than **56**, and would be the main product if the reaction was under full thermodynamic control, i.e. if all the steps were reversible.

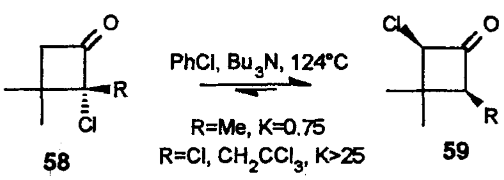
ered to be a steric effect, but could also be at least in part an electronic one.

Future research on this question could be interesting. Does the electronic effect of ligands, as expected, substantially change the polarity of metal-carbon bonds, and thereby influence isomerization equilibria[10] of alkyl transition metal complexes? This could lead to the design of more selective catalysts for processes such as hydroformylation, or the hydrocyanation of butadiene to adiponitrile [76].

5.3. Alkanes with Polyfunctional Carbon Atoms

Until now, this report has only treated alkanes bearing a single substituent per carbon atom, thus making sure that the only thermodynamic effects to be considered were due to electronegativity and possible steric hindering. In some more complicated cases, there are several substituents on a single carbon atom, and their interaction should be considered. However, the electronegativity effect does seem to be implied.

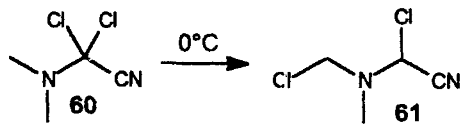
A first example is the cine rearrangement [77] of chlorinated cyclobutanones **58** to isomeric **59**. When R is an alkyl group, the chlorine substituent in **58** is in a favoured, tertiary alkyl, position. The reaction is equilibrated. When R is not electron-donating, such as Cl, CH_2CCl_3, the rearrangement to **59** is complete (scheme 27).



58 **59**

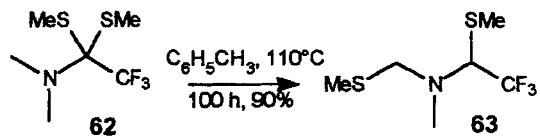
Scheme 27. Substituent effect on the *Cine* rearrangement.

Second, the covalent α,α-dichloro-amines **60** bearing an electron-withdrawing substituent Z isomerise quantitatively, at room temperature or above to give **61** (scheme 28). When we described this reaction some time ago [78], we did not speculate on its driving force. Now, it appears possible that it takes place because the chlorine migrates to a more electron-rich carbon atom. The reaction also occurs [79] with the *bis*-sulfide **62** to give **63** (scheme 29).



60 **61**

Scheme 28. Rearrangement of dichloroamines.



62 **63**

Scheme 29. Rearrangement of *bis*-sulfides.

6. Conclusions

Electronegativity can be a source of selectivity in organic synthesis for many reactions under thermodynamic control, in which a polar group becomes bonded to an alkyl group with a different degree of substitution than in the starting products. Many rearrangements have been studied only for their mechanism, i.e. their kinetic possibility, without a word on the thermodynamic reasons for which the rearrangement takes place. The question was: „How?", but not: „Why?".

Although the different stability of isomeric substituted alkanes has been described many times before, its significance for synthetic reactions has mostly gone unnoticed. Polar *substituents make stronger bonds to alkane carbon atoms with complementary polarity*: depending on the higher or lower electronegativity of the substituent, the attached carbon atom bears an induced partial positive or negative charge, and covalent species therefore have the same order of stability as the corresponding carbocations, or carbanions. This principle is documented by thermochemical data, and can be deduced from Pauling's formula for electronegativity.

The energy differences involved, of 1 to 7 kcal·mol^{-1} are small but significant, and are in the same range as many kinetic effects on activation energies such as steric hindrance in the transition state during asymmetric catalysis.

Although these electronegativity effects are only of qualitative predictive value, they do explain the selectivity of many known reactions, and, more importantly, provide means for the design of new selective syntheses.

The financial support of the Service de la Programmation de la Politique Scientifique (Belgium) is acknowledged.

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[10] Bennett et al. (ref. [59a,b]) have reported an interesting difference between the behavior of two alkyliridium complexes. In the $[RIrCl_2(CO)(PPh_3)_2]$ complexes shown in scheme 19, the equilibrium lies completely on the side of the n-isomer. In the dimeric chlorine-bridged octahedral complexes $[R_2Ir_2Cl_4(CO)_2]$, however, approximately equal amounts of *n*- and *iso*-propyl groups are present at equilibrium. The much lower *n*-preference could be due to the reduced electron density on the metal. However, it should be said that in the corresponding butyl complexes, the equilibrium mixture again contains almost entirely the *n*-butyl compound. In all the other cases we found in the literature, propyl and butyl complexes had a similar behavior, and give similar equilibrium constants (see esp. scheme 23). This suggests that the (surprising) difference between the propyl and butyl dimeric complexes needs to be reexamined.

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